CHAPTER 18 · DERIVATIONS

ELECTROMAGNETIC INDUCTION

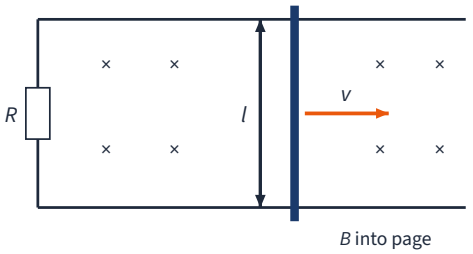
5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 E.M.F. IN A MOVING STRAIGHT CONDUCTOR

SYLLABUS 18.2(C)

RESULT $\varepsilon = Blv \sin\theta$

1. A rod of length l moves with velocity v at angle θ to the field B (rod perpendicular to B).
2. In time dt it sweeps an area $l(v \sin\theta)dt$ across the field lines.
3. Flux cut $d\Phi = Blv \sin\theta\, dt$.
4. Faraday's law: $\boldsymbol{\varepsilon = \frac{d\Phi}{dt} = Blv \sin\theta}$.



Method 2 (charge moving in a magnetic field): each free electron in the rod feels qvB along the rod; charges separate until qE = qvB, so ε = El = Blv.

2 E.M.F. IN A ROTATING COIL

SYLLABUS 18.2(D)

RESULT $\varepsilon = NBA\omega \sin\omega t$

1. A coil of N turns and area A rotates at angular velocity ω about an axis perpendicular to B.
2. Starting with its plane perpendicular to B, the angle between B and the normal is $\theta = \omega t$.
3. Flux linkage $N\Phi = NBA \cos\omega t$.
4. $\boldsymbol{\varepsilon = -\frac{d(N\Phi)}{dt} = NBA\omega \sin\omega t}$; peak value $\varepsilon_0 = NBA\omega$.

3 SELF-INDUCTANCE OF A SOLENOID

SYLLABUS 18.3(D)

RESULT $L = \frac{\mu_0 N^2 A}{l}$

1. Long solenoid: N turns, length l, area A, current I: $B = \frac{\mu_0 NI}{l}$.
2. Flux through each turn $\Phi = BA = \frac{\mu_0 NIA}{l}$.
3. Flux linkage $N\Phi = \frac{\mu_0 N^2 AI}{l}$.
4. $LI = N\Phi$, so $\boldsymbol{L = \frac{\mu_0 N^2 A}{l}}$.

4 BACK E.M.F. IN A COIL

SYLLABUS 18.3

RESULT $\varepsilon = -L\dfrac{dI}{dt}$

1. Flux linkage in the coil is proportional to the current: $N\Phi = LI$.
2. Faraday's law: $\varepsilon = -\dfrac{d(N\Phi)}{dt}$.
3. Differentiate: $\varepsilon = -\dfrac{d(LI)}{dt}$, and L is constant, so $\boldsymbol{\varepsilon = -L\dfrac{dI}{dt}}$.
4. The minus sign (Lenz's law): the back e.m.f. opposes the change in current.

5 MUTUAL INDUCTANCE OF TWO COAXIAL SOLENOIDS

SYLLABUS 18.5(D)

RESULT $M = \dfrac{\mu_0 N_p N_s A}{l_p}$

1. Primary solenoid (N_p turns, length l_p) carries I_p: $B = \dfrac{\mu_0 N_p I_p}{l_p}$.
2. The secondary (N_s turns) is wound coaxially with the same area A: $\Phi_s = BA$.
3. $N_s\Phi_s = \dfrac{\mu_0 N_p N_s A I_p}{l_p}$.
4. $MI_p = N_s\Phi_s$, so $\boldsymbol{M = \dfrac{\mu_0 N_p N_s A}{l_p}}$.